

Research Statement

Structural Methods for Mixed Model/Data Digital Twin Engineering

Ismail AISSAOUI | State Engineer in AI

1 Introduction and Motivation

Modern engineering digital twins are built upon large-scale physical models derived from first principles. However, the complexity of these models often leads to significant discrepancies when compared to real-world data, originating from parameter uncertainty, sensor drift, or structural modeling deficiencies. This research statement proposes a framework to address these "reality gaps" using structural analysis of differential-algebraic equation (DAE) systems, enabling the precise localization and correction of model/data mismatches.

2 Current Research: Physics-Informed Digital Twins

In my current work at Sonatrach, I have developed a generative surrogate model for gas turbine monitoring using **Mamba-SSM** and **xLSTM** architectures. My contribution lies in the design of **Physics-Informed Neural Network (PINN)** loss functions that enforce thermodynamic consistency. This work has highlighted a critical challenge: when a discrepancy occurs, is it due to the data, the parameters, or the fundamental physical equations? My experience has demonstrated the need for a formal structural diagnostic layer that precedes or accompanies the data assimilation process.

3 Proposed Research Directions

Building upon the objectives of the **Structural Methods** project at Inria Rennes, I propose to investigate the following axes:

3.1 1. Graph-Based Structural Analysis for DAEs

I intend to leverage my background in **Graph Neural Networks (GNNs)** to develop scalable algorithms for the structural analysis of large-scale DAE systems. By representing equations and variables as bipartite graphs, I will explore the computation of Minimal Structurally Overdetermined (MSO) subsystems. These subsystems act as "parity spaces," providing a mathematical foundation for detecting inconsistencies without requiring the full reachability analysis of the system.

3.2 2. Localizing Model/Data Mismatches

The core of my research will focus on using the residuals generated by solved MSO subsystems to localize deficiencies. By performing statistical analysis on these residuals, I aim to distinguish between sensor errors (data-driven) and structural deficiencies in the underlying Modelica or port-Hamiltonian models. This targeted diagnosis is essential for the efficient maintenance of complex industrial twins.

3.3 3. Scalability for Large Sparse Systems

Real-world industrial systems can contain thousands of equations. I propose to optimize the structural analysis workflow for large sparse systems by leveraging high-performance graph algorithms. The goal is to ensure that the diagnostic tools remain efficient for real-time or near-real-time synchronization within the digital twin lifecycle.

4 Alignment with the EDT Program

My background in relational modeling and industrial AI implementation aligns with the interdisciplinary goals of the **EDT program**. I am eager to contribute to the **Artemis platform** by providing a robust structural diagnostic layer that ensures the "truthfulness" of the hybrid models used across the consortium.

5 Conclusion

The goal of my PhD will be to transform digital twin engineering from ad-hoc data assimilation to a rigorous, structurally-aware diagnostic process. By bridging the gap between graph theory, DAE structural analysis, and hybrid modeling, I aim to provide the tools necessary for the next generation of reliable and self-diagnosing digital twins.